

Quantum Computing: A Long-Term Investment Thesis

**Based on institutional research, company filings and
market data**

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Executive Summary

Quantum computing is an emerging technology with the potential to solve selected problems that are beyond the practical capabilities of classical computers. Its most promising applications include drug discovery, molecular and materials simulation, financial modelling, logistics optimization, cybersecurity and advanced industrial processes. Quantum systems are unlikely to replace classical computers, instead, they are expected to operate alongside high-performance computing and artificial intelligence within hybrid computing environments.

The sector remains at an early stage of commercialization. Current systems face major technical constraints, including unstable qubits, high error rates, limited connectivity, complex manufacturing and demanding cooling requirements. The key milestone will be the development of scalable, fault-tolerant quantum computers capable of delivering repeatable economic value. Meaningful adoption is therefore expected to develop gradually, with early commercial use cases emerging during the late 2020s and broader adoption more likely during the 2030s.

Despite these uncertainties, the commercial ecosystem is expanding. Quantum-computing companies generated more than \$1 billion of global revenue in 2025, potentially rising to approximately \$4.4 billion by 2028. Investment in quantum-technology start-ups also reached \$12.6 billion in 2025, indicating increasing private-sector confidence, although funding is becoming concentrated among a relatively small group of leading companies.

From an investment perspective, the public market consists of two distinct groups. Pure-play developers such as IonQ, Quantinuum, D-Wave and Rigetti offer the greatest direct exposure to quantum commercialization. However, they currently generate limited revenue, report substantial operating losses and consume significant amounts of cash. Their valuations reflect expectations of future technological leadership rather than current earnings, creating high execution, dilution and valuation risk.

Established enabling suppliers such as Keysight Technologies, Coherent and Oxford Instruments provide a more balanced risk-return profile. These companies generate meaningful revenue, positive operating margins and cash flow from existing test, measurement, photonics and communications businesses, while retaining indirect exposure to quantum-sector growth.

The principal investment opportunity is the possibility that quantum computing develops into a foundational enterprise technology and that selected companies establish dominant positions within the ecosystem. The main risk is that the technology succeeds while many current public companies fail to capture the resulting economic value.

Overall, quantum computing is an attractive but highly speculative long-term investment theme. It should be treated as a small satellite allocation rather than a core portfolio holding. For a growth-oriented investor with a 10–15-year horizon, an allocation of approximately 3–5% may be appropriate, preferably diversified across selected pure-play developers and established enabling suppliers. Continued monitoring of technological milestones, revenue growth, cash burn, customer adoption and competitive positioning will be essential.

Introduction to Quantum Computing

What is Quantum Computing

Quantum computing is a new approach to calculation that uses principles of fundamental physics to solve extremely complex problems very quickly. Quantum computers, a fundamentally different approach from digital computers, are poised to take computing to a whole new level. Quantum computers have the potential to solve very complex statistical problems that are well beyond the limits of today's computers, across a range of industries and applications, including finance, transportation, pharmaceuticals, and green technology.

Classical computing, the technology that powers your laptop and smartphone, is built on bits. A bit is a unit of information that can store either a zero or a one. By contrast, quantum computing is built on quantum bits, or qubits, which can store both zeros and ones. Qubits can represent any combination of both zero and one simultaneously; this is called superposition, and it is a basic feature of any quantum state.

When a classical computer uses multiple variables to solve a problem, it must conduct a new calculation every time one of those variables changes. Each calculation is a single path to a single result. Quantum computers, on the other hand, can explore many paths in parallel through superposition.

Quantum Computing capabilities

Quantum computers don't deliver just one clear answer like classical computers do, but a range of possible answers. For calculations that are limited in scope, classical computers are still the preferred tools. But for very complex problems, quantum computers can save time by narrowing down the range of possible answers.

Progress in quantum computing, however, is expected to remain a long-term race to achieve the quantum volume that's required to quickly solve real-world problems. According to McKinsey's conversations with tech executives, investors, and academics in quantum

computing, 72% believe we'll see a fully fault-tolerant quantum computer by 2035. The remaining 28% think this milestone won't be reached until 2040 or later.

Main Breakthroughs

In 2024, Google unveiled an experimental quantum computer that, in five minutes' time, could perform a calculation that would take most supercomputers ten septillion years to finish which is longer than the age of the known universe. Google's quantum chip, called Willow, is primarily suited for research and specialized fields. In February 2025, Microsoft announced the discovery of a new state of matter that it says will support a quantum computing breakthrough. After 17 years of physics research, the tech giant unveiled the Majorana 1 quantum chip. Microsoft says the chip could be used to accelerate drug discovery, battery development, and the race toward AI dominance.

Currently, organizations are mainly using five qubit technologies in their attempts to build a scalable, universal quantum computer. According to the McKinsey data in the chart below, superconducting circuits have received the highest overall investment among the five qubit technologies, totaling \$2,732 million.

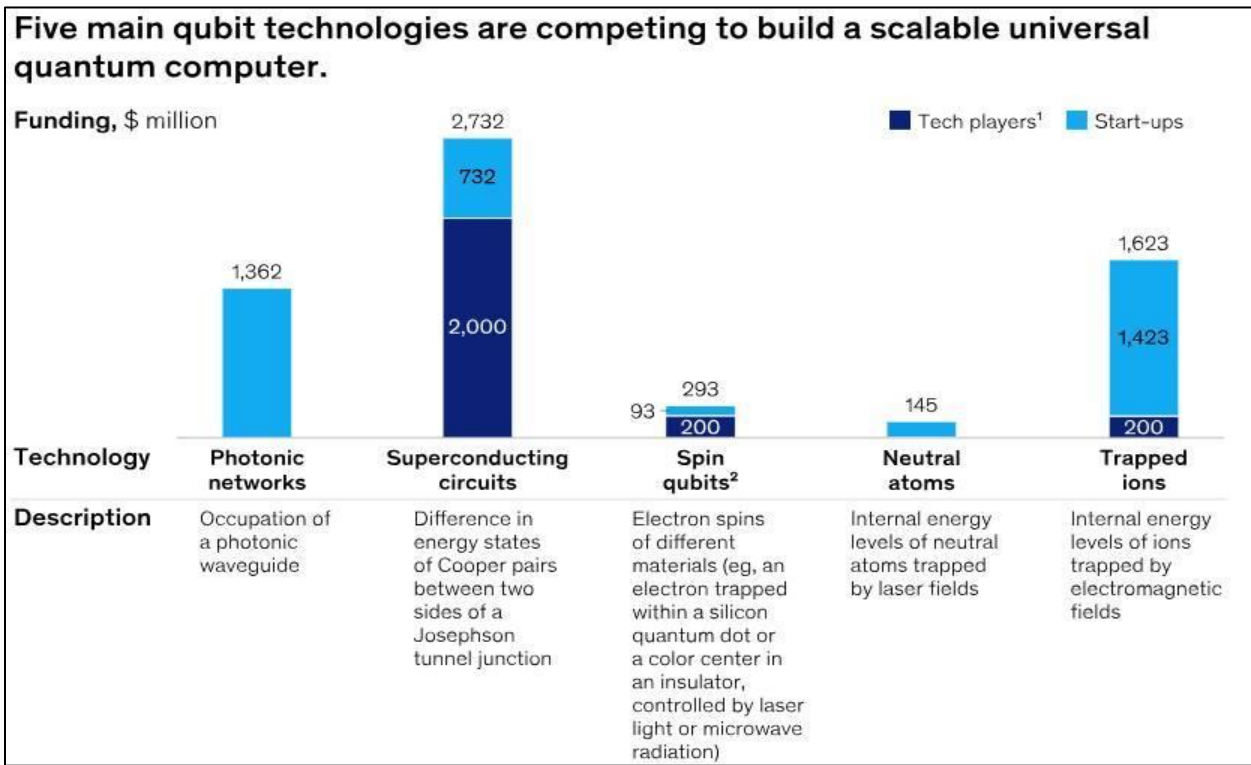


Chart 1: Competing Quantum Technologies (McKinsey & Company)

Quantum & AI

Quantum computing and AI have a potentially symbiotic relationship, and they stand to advance each other's capabilities and respective fields.

Potential Benefits of AI from Quantum Computing:

- Accelerating AI processes and training models
- Enhanced computational power
- More efficient problem-solving
- Improved learning capabilities
- Parallel processing
- Easier handling of complex data structures

Potential Benefits of Quantum Computing from AI

- Error correction
- Noise reduction
- Quantum-algorithm design and optimization
- Resource management
- Simulation and emulation
- Benchmarking and performance analysis
- Quantum machine learning

Challenges

The major challenge in the advancement of quantum computing is the volatility of qubits due to their dual property. When a qubit changes its status, inputs can be lost or altered, which impacts the accuracy of the results. Besides this, the technology faces the following obstacles:

- Maintain accuracy and reliability greater than 99.99% at scale
- Interaction speed between qubits
- Multiqubit networking
- Cooling power and environmental control
- Manufacturability

Future Potential

Economic Value

According to McKinsey & Company, quantum computing companies generated more than \$1 billion in revenue worldwide in 2025, and that could grow to as much as \$4.4 billion by

2028. However, this industry, could create up to \$2.7 trillion of economic value worldwide by 2035.

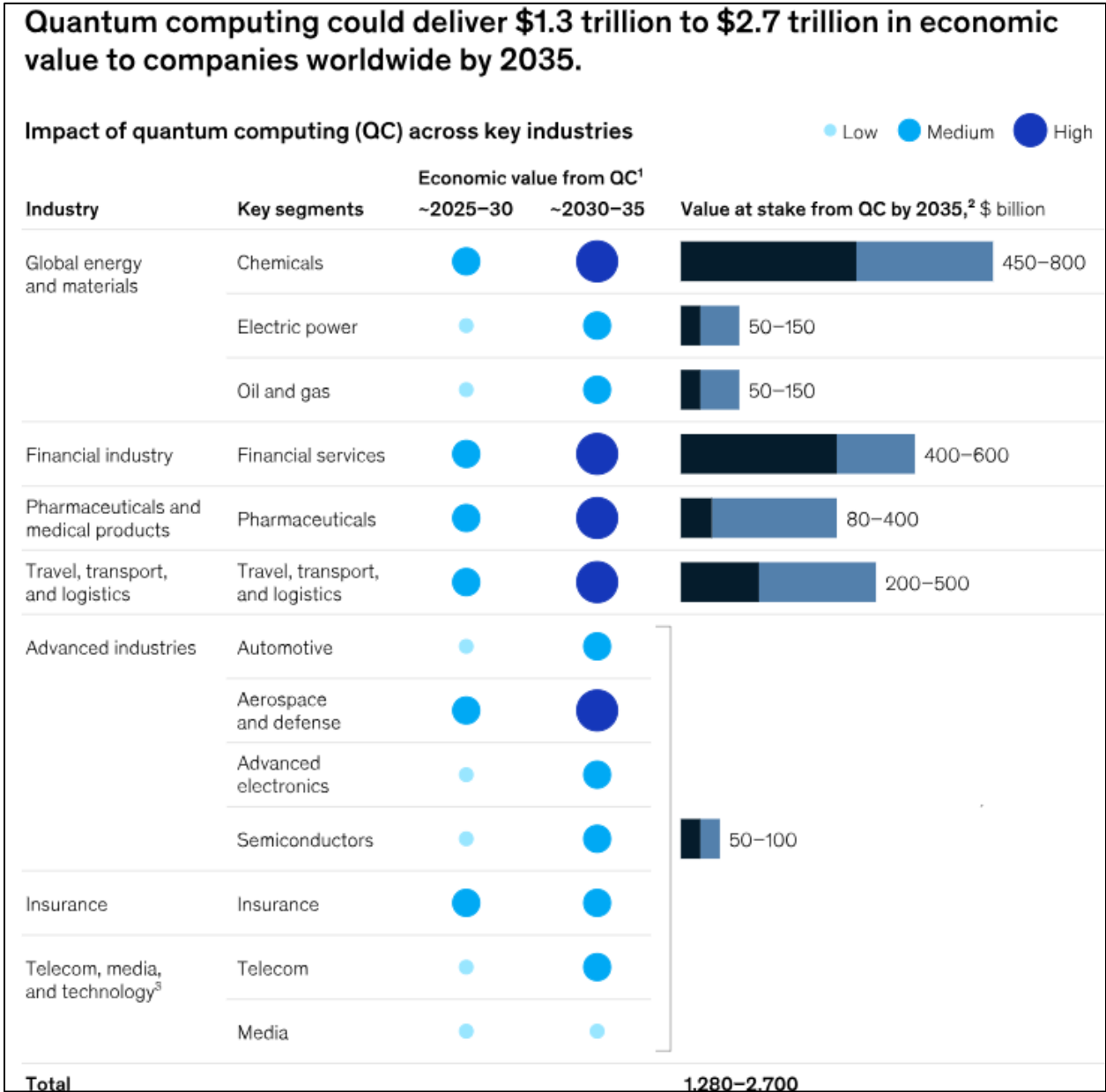


Chart 2: Quantum Computing Economic Value (McKinsey & Company)

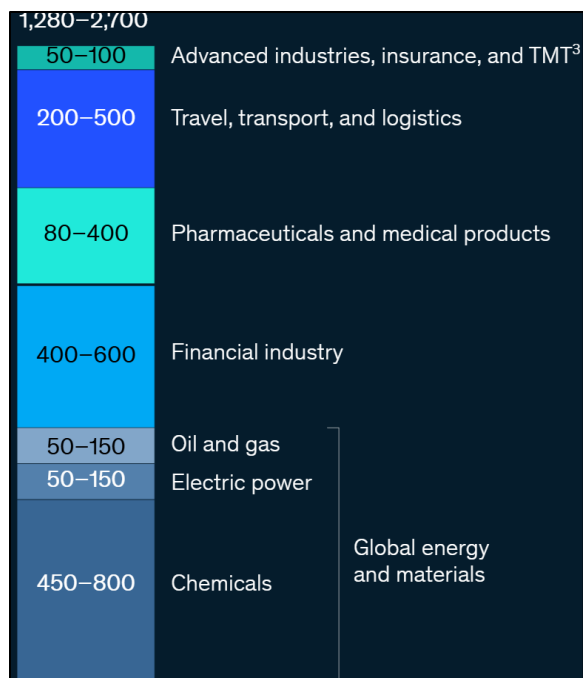


Chart 3: Value at stake from QC by 2035 by key segment in billion USD (McKinsey & Company)

Geographic distribution

Research shows that European companies are leading when it comes to advancing quantum computing and that 72 percent of quantum computing use is at companies that are majority-owned by private entities.

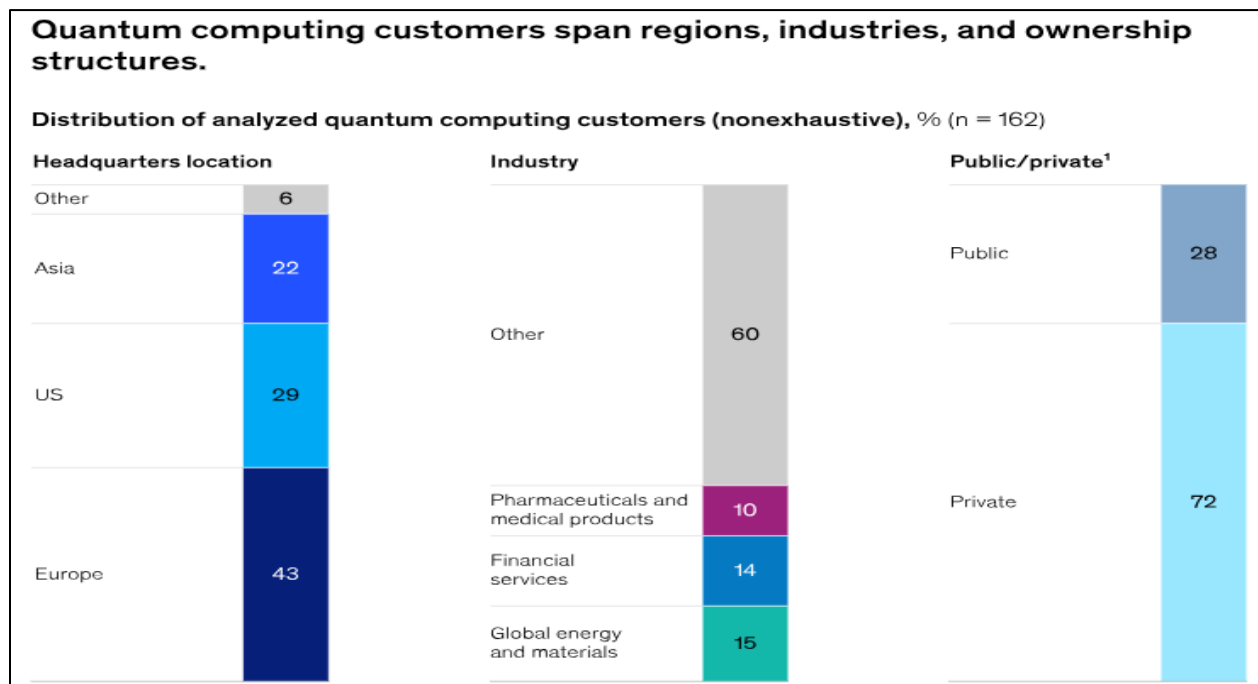


Chart 4: Quantum computing company analysis(McKinsey & Company)

Spending and Investment

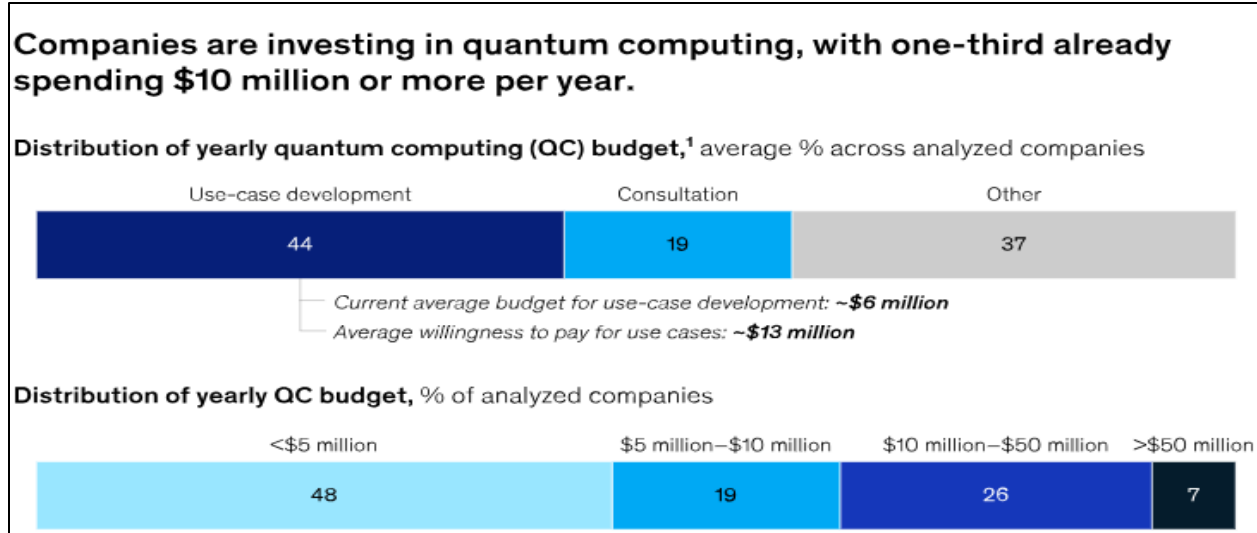


Chart 5: Quantum computing company spending(McKinsey & Company)

Companies making the fastest progress with quantum computing are those that pair technical experimentation with clear economic hypotheses and defined delivery road maps. JPMorgan Chase, for example, has an internal team of scientists building quantum algorithms and applications to address business use cases in AI, portfolio optimization, and cryptography.

Quantum computing is taking off in a few key industries:

- Chemicals and life sciences
- Travel, transport, and logistics
- Financial services

Investment in quantum technology start-ups reached \$12.6 billion in 2025, 6.3 times higher than in 2024. A full 90 percent of this investment went to quantum computing start-ups.

Private investment in quantum technology start-ups is booming, driven by capital market deals.

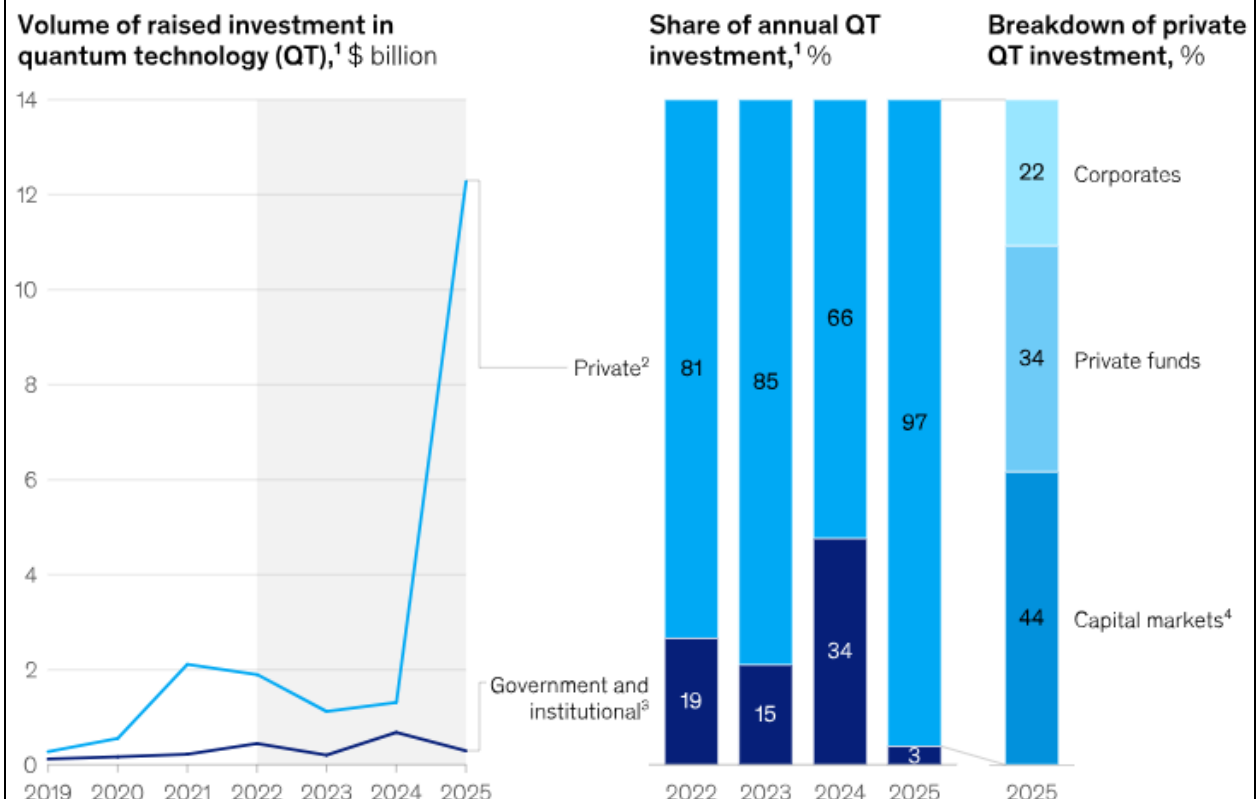


Chart 6: Investment distribution (McKinsey & Company)

A marked change from 2025 is the shift from public to private investment. In 2024, one-third of investment in quantum technology start-ups came from public sources such as governments, sovereign wealth funds, and universities. In 2025, just 3% did.

Investment in the quantum industry became increasingly concentrated in 2025, with around 60% of total funding going to the ten largest deals. This reflects a maturing market where valuations are rising, access to talent and technological progress is concentrated among a few leading companies, and the cost of developing quantum technologies continues to increase. At the same time, M&A activity accelerated significantly, further signaling industry consolidation.

Technology Maturity

The quantum technology industry is transitioning from a research-driven ecosystem to a more commercial and scalable technology market, encompassing quantum hardware, software, and services. The global **internal** quantum market (market for the technology itself, including hardware, software, and services) is projected to reach \$60–100 billion by 2035,

with quantum computing expected to account for \$43–71 billion of that total. The lower-end forecast for quantum computing was raised from \$28 billion to \$43 billion, reflecting increased willingness among companies to invest in quantum technologies. (McKinsey & Company)

While technical progress is accelerating, uncertainty remains around development roadmaps, fault-tolerance timelines, and the timing of meaningful returns on investment. In the near term, the most practical path to value is through hybrid computing, where quantum systems work alongside classical computing and AI to solve the most complex parts of a problem rather than replacing traditional systems entirely.

Investment Assessment

As mentioned in the previous chapter, the quantum computing market revenue is expected to grow significantly in the next decades:

Period	Beginning Revenue	Ending Revenue	CAGR
2025 - 2028	\$1B	\$4.4B	63.9%
2028-2035	\$4.4B	\$97B	55.6%
2035-2040	\$97B	\$198B	15.3%
2025-2040	\$1B	\$198B	42.3%

Table1:**Total** Quantum Technology market revenue over the years (McKinsey & Company)

In the case of the Quantum Computing sector, a single industry average index of Exchange Traded fund would be misleading because the investable universe contains two fundamentally different groups:

1. Pure play quantum developers
 - a. IonQ, Quantinuum, D-Wave and Rigetti (Appendix A)
2. Established enabling suppliers
 - a. Keysight, Oxford Instruments and Coherent (Appendix B)

Pure play quantum developers

Metric	IonQ	Quantinuum	D-Wave	Rigetti
Market Capitalization	\$12.7B	\$1.8B	\$6.19B	\$4.74B
Enterprise Value	\$10.78B	-\$447.65M	\$5.65B	\$4.33B
Price/Sales	57.82	110.33	458.8	457.55
Enterprise Value/Revenue	57.6	3.46	454.22	431.83
Revenue	\$187.12M	\$17.08M	\$12.44M	\$10.02M
Operating Margin	-401.75%	-1473%	-1914.87%	-589.79%
EBITDA	-\$662.37M	-\$215.53M	-\$138.81M	-\$80.02M
Total Debt/Equity	0.61%	1.61%	4.17%	1.16%
Levered Free Cash Flow	-\$91.36M	NA	-\$63.51M	-\$6.21M
Operating Cash Flow	-\$401.19M	-\$190.44M	-\$97.66M	-\$61.11M
Beta	3.23	NA	2.1	1.96

Table 2: Financial Data of pure play quantum developers (Yahoo Finance, July 2026)

The financial profiles of pure-play quantum computing companies highlight the central paradox of quantum investing: these firms command multi-billion-dollar market capitalizations despite generating only limited revenue and incurring substantial operating losses. Investors are clearly not valuing these businesses based on their current earnings power. Instead, they are pricing them as long-duration options on the future commercialization and widespread adoption of quantum computing.

Valuation multiples are exceptionally elevated relative to conventional technology companies. At these levels, incremental growth is unlikely to be sufficient. To justify their current valuations, these companies must ultimately deliver extraordinary revenue expansion and establish commercially viable business models at scale.

Persistently negative operating margins provide the strongest evidence that the industry remains in an early development phase. These losses are driven by the significant investments required to recruit highly specialized scientists and engineers, build quantum hardware infrastructure, and fund software, algorithm, and research development. Negative operating cash flows further reinforce the developmental nature of the sector and create three key risks for investors: funding risk, dilution risk, and execution risk.

The generally low levels of debt relative to equity suggest that capital raising is primarily achieved through equity issuance rather than borrowing. While this reduces financial leverage risk, it increases the likelihood of shareholder dilution as companies continue to

fund their operations and technological development before reaching sustainable profitability.

Beta of close to 2 and above indicates that the stocks have historically moved much more sharply than the broader equity market. This is consistent with their economic profile as most of their value lies many years into the future.

These numbers do not describe financially mature technologies, rather publicly traded R&D platforms whose valuations are based on the probability of future growth and return.

Factor	Assessment
Growth potential	Very High
Current profitability	Very low
Revenue	Low
Balance Sheet risk	Moderate: Low debt but high cash burn
Capital Intensity	Very High: Significant investments required
Dilution risk	High
Valuation risk	Very High
Technological upside	Very High
Suitable portfolio role	Small speculative satellite allocation
Appropriate holding period	10-15 years

Table 3: Investment Assessment of pure play quantum developers

Established enabling suppliers

Metric	Keysight	Oxford Instruments	Coherent
Market Capitalization	\$53.68B	£1.56B	\$55.8B
Enterprise Value	\$54.03B	£1.5B	\$56.84B
Price/Sales	8.9	3.87	7.79
Enterprise Value/Revenue	8.87	3.55	8.61
Revenue	\$6.09B	£423.2M	\$6.6B
Operating Margin	18.81%	18.55%	13.57%
EBITDA	\$1.43B	£81.8M	\$1.31B
Total Debt/Equity	43.64%	13.08%	31.09%
Levered Free Cash Flow	\$1.09B	£71.47M	-\$197.63M
Operating Cash Flow	\$1.49B	£54.4M	\$140.34M
Beta	1.21	1.11	2.04

Table 4: Financial Data of established enabling suppliers (Yahoo Finance, July 2026)

Unlike the pure-play quantum computing companies, Keysight, Oxford Instruments, and Coherent are established enabling-technology providers with meaningful revenue bases,

positive profitability, and proven cash-generation capabilities. While all three companies have exposure to quantum computing through instrumentation, photonics, lasers, cryogenics, and test equipment, investors are primarily valuing them based on their existing businesses rather than on speculative future quantum market adoption.

Valuation multiples appear elevated relative to traditional industrial companies but remain well below the levels typically observed among pure-play quantum firms. Price-to-sales ratios ranging from approximately 4x to 9x suggest that investors are assigning value to sustainable growth opportunities and strategic exposure to secular technology trends such as AI, semiconductors, photonics, and quantum technologies, while still anchoring valuations to established earnings and cash flow generation. This creates a more balanced risk-reward profile than that seen in pre-commercial quantum companies.

Operating margins between 13.6% and 18.8% indicate healthy underlying business economics and demonstrate that these companies have already achieved commercial scale. Unlike pure-play quantum firms that are investing heavily ahead of revenue generation, these businesses can fund a significant portion of their research and development efforts through internally generated profits. Positive EBITDA across all three companies further reinforces their operational maturity and financial resilience.

Cash flow generation is a key differentiator. Keysight and Oxford Instruments generate positive operating and free cash flow, providing flexibility to invest in new technologies, pursue acquisitions, and return capital to shareholders. Coherent generates positive operating cash flow but currently exhibits negative leveraged free cash flow, suggesting that capital expenditures, investments, or financing-related obligations are temporarily consuming more cash than the business is producing after those expenditures. Nevertheless, the company remains fundamentally different from cash-burning development-stage quantum companies because it continues to generate cash from its core operations.

Balance sheet risk appears manageable across the peer group. Debt-to-equity ratios range from 13% at Oxford Instruments to 44% at Keysight, levels that are generally consistent with mature technology and industrial businesses. The ability to support operations from recurring cash flows reduces dependence on external capital markets and significantly lowers dilution risk compared with early-stage quantum computing firms that frequently issue equity to fund ongoing development.

From a market-risk perspective, Keysight and Oxford Instruments exhibit betas close to 1, indicating share price movements broadly in line with the wider equity market. Coherent's beta of approximately 2 suggests materially higher volatility and sensitivity to economic

cycles, technology spending, and investor sentiment. However, unlike pure-play quantum companies where much of the valuation depends on future technological breakthroughs, these firms derive the majority of their value from existing commercial operations, making their investment profiles considerably less speculative.

Overall, the investment case for Keysight, Oxford Instruments, and Coherent is not based on a binary outcome regarding quantum computing commercialization. Instead, they offer investors exposure to the growth of the quantum ecosystem through a "picks-and-shovels" approach. As demand for quantum research, advanced semiconductors, photonics, and high-performance computing infrastructure increases, these companies are positioned to benefit regardless of which quantum computing platform or developer ultimately emerges as the long-term industry winner.

Factor	Assessment
Growth potential	Moderate to High
Current profitability	Healthy / positive
Revenue	Established revenue base, unlike pure-play quantum firms
Balance Sheet risk	Low to moderate/ manageable leverage overall
Capital Intensity	Moderate
Dilution risk	Low/ less dependent on equity issuance to fund operations
Valuation risk	Moderate/ multiples are elevated but supported by existing businesses
Technological upside	High, but indirect exposure to quantum commercialization
Suitable portfolio role	Established technology exposure with quantum optionality
Appropriate holding period	5-10 years

Table 5: Investment Assessment of established enabling suppliers

Overall conclusion

Quantum computing offers significant long-term potential, with estimates of up to \$2.7 trillion in economic value by 2035. However, the investment case remains highly uncertain. Commercial adoption is expected to occur gradually through hybrid computing and quantum-as-a-service models, with meaningful growth likely emerging in the late 2020s and broader commercialization in the 2030s.

Publicly listed pure-play quantum companies remain early-stage investments. Revenues are limited, cash burn is high, and shareholders face dilution, execution, and technology risks. As a result, traditional valuation metrics are less useful than indicators such as cash runway, customer adoption, technological progress, and intellectual property strength.

Established suppliers provide a more balanced way to gain exposure to the sector. They benefit from quantum-related spending while generating profits from broader businesses.

The key investment takeaway is that belief in quantum computing does not guarantee success for today's quantum companies. Diversification across hardware developers and enabling technologies is therefore preferable to betting on a single winner.

For most investors, quantum computing is best viewed as a small speculative allocation (around 3–5% of a portfolio) with a long-term horizon of 10-15 years. The sector offers extraordinary technological upside but currently suffers from weak economics, limited visibility, and demanding valuations. A diversified, staged investment approach remains the most prudent strategy.

Bibliography

McKinsey & Company, 2026. *McKinsey Quantum Technology Monitor 2026*. New York: McKinsey & Company.

McKinsey & Company, 2025. *The Rise of Quantum Computing*. New York: McKinsey & Company.

McKinsey & Company, 2025. *The Year of Quantum: From Concept to Reality in 2025—Quantum Technology Monitor 2025*. New York: McKinsey & Company.

McKinsey & Company, 2025. *What Is Quantum Computing?* McKinsey Explainers. New York: McKinsey & Company.

Yahoo Finance (2026). Company Statistics and Market Data for IonQ, Quantinuum, D-Wave Quantum, Rigetti Computing, Keysight Technologies, Oxford Instruments and Coherent. Accessed July 2026.

Appendix A

IonQ's long-term vision is to build quantum computers with 2 million qubits by 2030, enabling breakthroughs in areas such as drug discovery, materials science, finance, logistics, cybersecurity, and defense. In parallel, IonQ is investing in quantum networking, sensing, security, and space infrastructure, positioning itself as a key contributor to the future quantum internet.

As the world's largest integrated quantum company, Quantinuum is leading the development of the most powerful quantum computers and the most advanced quantum software solutions. With our full-stack technology and world-class scientists, we are rapidly scaling quantum computing to solve tomorrow's biggest challenges.

D-Wave is a pioneer in quantum computing and was the world's first commercial supplier of quantum computers. The company is unique in offering a dual-platform approach, providing both quantum annealing and gate-model quantum computing technologies through its systems, software, and services. D-Wave's mission is to help organizations realize practical quantum value today by delivering enterprise-grade quantum solutions both on-premises and through its Leap™ quantum cloud platform, which offers 99.9% availability and uptime.

Rigetti Computing is an integrated systems company. We build quantum computers and the superconducting quantum processors that power them. Through our Quantum Cloud Services (QCS) platform, our machines can be integrated into any public, private or hybrid cloud.

Appendix B

Keysight Technologies, Inc. is a global company headquartered in Santa Rosa, CA, that manufactures hardware and software for engineering workflows across design, test, and emulation. It serves industries including communications, aerospace and defense, AI/data center networking, automotive, semiconductors, digital healthcare, quantum computing, and energy.

Oxford Instruments is a leading provider of high technology products and services to the world's leading companies and scientific research communities. Our businesses image, analyze and manipulate materials down to the atomic and molecular level, enabling our customers to accelerate their R&D, increase productivity and make new scientific discoveries.

Coherent is the global photonics leader. We harness photons to drive innovation. Industry leaders in the datacenter, communications, and industrial markets rely on Coherent's world-leading technology to fuel their own innovation and growth. Founded in 1971 and operating in more than 20 countries, Coherent brings the industry's broadest, deepest technology stack; unmatched supply chain resilience; and global scale to help its customers solve their toughest technology challenges.